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Fluid sampler

Abstract

An apparatus is provided for obtaining fluid samples with the apparatus having a tubular body with a top and a bottom. A spring operated plunger shaft extends axially from the top to the bottom of the tubular body. A sample volume cap is attached to an end of the spring operated plunger shaft. A plunger is attached to the spring operated plunger shaft. A fusible link is used to maintain the spring operated plunger shaft in a first position. Upon activation of the fusible link, axial motion of the spring operated plunger shaft draws fluid into the tubular body trapping the fluid between the plunger and the sample volume cap inside the tubular body.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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Background/Summary

CROSS REFERENCE TO OTHER PATENT APPLICATIONS

(1) None.

BACKGROUND OF THE INVENTION

1) Field of the Invention

(2) The present invention is directed to a remotely operated fluid sampler.

2) Description of the Related Art

(3) In oceanographic research and waterway management, there is a need to collect water samples. In many circumstances when the number of required samples is small or when access by an operator is convenient, manual collection methods, including pipettes, syringes, and small containers may be appropriate. Water sampling systems of these types are well known in the art and are commercially available. However, numerous circumstances exist where the water to be sampled is not easily accessible, such as in deep water and wells. Also, the timing for obtaining the sample is inconvenient, such as when periodic samples are collected over a long period of time in a remote area.

(4) Water sampling devices of this type are not generally available and although examples of systems designed for these applications exist; the systems do not provide sufficient capabilities. One solution employed in the oceanographic research field uses evacuated bottles that are subsequently filled with water samples. The bottles are submerged, and when placed in the location where a sample is desired, the bottles are individually opened with a valve and water is drawn in through the valve.

(5) Multiple bottles are clustered with individual valves to fill discrete bottles at desired sample timing. This solution can be modified to include pumps to force water into the chambers. Designs employing this approach include valve systems that are complex and costly. Furthermore, the bottles must withstand high pressures prior to filling if employed in environments at high ambient pressure such as deep water environments.

(6) A device is needed that can be remotely deployed and subsequently activated to collect multiple samples of water (or other fluid) in which it has been placed, on a sampling schedule that may extend over a long period of time. The solution should be comparatively low cost, mechanically simple, suitable for employment at high ambient pressures, and suitable for remote or autonomous actuation.

SUMMARY OF THE INVENTION

(7) The invention is a fluid sampling system that can be remotely deployed and activated to collect multiple samples of water (or other fluid) in which the system is placed. The system can be placed

on an extendable or specified sampling schedule.

(8) The system includes a remotely actuated water sampler with an integrated storage reservoir and seal. The water sampler is modular such that larger numbers can be integrated into a larger sampler system to collect multiple water samples on a specified schedule.

(9) The sampler includes a body having a spring retention section, a sample volume section, and a sample strainer section. The bottom of the spring retention section is attached to the top of the sample volume section and the bottom of the sample volume section is connected to the top of the sample strainer section.

(10) A plunger shaft having a first end and a second end is positioned in the body and extends axially from the top of the spring retention section, through the sample volume section, and to the bottom of the sample strainer section. A spring is disposed around the first end of the plunger shaft. A sample volume cap is attached to the second end of the plunger shaft with a plunger attached to the plunger shaft. The plunger is located between the sample volume section and the sample strainer section.

(11) In another embodiment of the invention, a fluid sampler module has a tubular body with a top and a bottom. A spring operated plunger shaft extends axially from the top to the bottom of the tubular body. A sample volume cap is attached to an end of the spring operated plunger shaft with a plunger attached to the spring operated plunger shaft. Axial motion of the plunger shaft draws fluid into the tubular body with the result of trapping the fluid between the plunger and the sample volume cap.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other objects, features and advantages of the present invention will become apparent upon reference to the following description of the preferred embodiments and to the drawings, wherein corresponding reference characters indicate corresponding parts throughout the several views of the drawings and wherein:

(2) FIG. 1 is a cutaway view of a sampler module;

(3) FIG. 2 is a side view of a fusible link on a sampler module; and

(4) FIG. 3 is a perspective view of a fluid sampling system having a plurality of sampler modules.

DETAILED DESCRIPTION OF THE INVENTION

(5) Referring to FIG. 1, a sampler module **100** includes a body **102**. The body **102** includes a plurality of tube sections and a spring retention section **104**, a sample volume section **106** and a sample strainer section **108**. The bottom of the spring retention section **104** is attached to the top of the sample volume section **106**. In some embodiments, the top of the sample volume section **106** includes threads **110** so that the spring retention section **104** can be threadedly engaged with the sample volume section.

(6) The bottom of the sample volume section **106** is connected to the top of the sample strainer section **108**. In some embodiments, the sample volume section **106** and the sample strainer section **108** can be formed together by additive manufacturing. The sample strainer section **108** is capable of allowing fluid to pass into and out of the strainer section.

(7) A plunger shaft **112** having a first end **114** and a second end **116** is installed in the body **102** and extends axially from the top of the spring retention section **104**, through the sample volume section **106**, to the bottom of the sample strainer section **108**. At the first end **114**, a spring **118** is provided around the plunger shaft **112** and within the spring retention section **104**. The top of the sample volume section **106** includes an aperture **120** for the plunger shaft **112** to pass therethrough. The spring **118** rests on the top of the sample volume section **106** and is held in place by a disk **122**, which is attached to the plunger shaft **112** by a nut **124** or other appropriate fastener.

(8) At the second end **116**, a sample volume cap **126** is positioned on the plunger shaft **112**, within the sample strainer section **108**. A plunger **128** is provided on the plunger shaft **112** between the sample volume section **106** and the sample strainer section **108**.

(9) In a first axial position shown in FIG. **1**, the spring **118** is compressed axially between the top of the sample volume section **106** and the bottom of the shaft push disk **122**, holding the plunger shaft **112** in position with the sample volume cap **126** extended into the sample strainer section **108**. In this first position, the plunger **128** is at the bottom of the sample volume section **106**.

(10) The plunger shaft **112** is held in the first axial position by pins **130** that protrude from the periphery of the disk **122**. Referring to FIG. **2**, the pins **130** pass through a J-slot **132** located at the top of the spring retention section **104**. A fusible link **134** holds the plunger shaft **112** and disk **122** in place by preventing motion of the pins **130** inside the J-slot **132**. Tensile loads from the spring **118** are mitigated by sloping of the J-slot **132** where the pins **130** are received and the associated friction forces between the pins **130** and the J-slot.

(11) In operation, the sample strainer section **108** extends into the fluid environment to be sampled. When a sample from the environment is desired, an electrical current is passed through the fusible link **134**, causing it to burn and fail. When the fusible link **134** fails, the pins **130** are allowed to move in the J-slot **132**. Force from the spring **118** against the disk **122** causes the disk to rotate with the pins **130** in the J-slot **132** until the pins are free and the disk can move axially. The spring then forces the disk **122**, plunger shaft **112**, and plunger **128** to a second axial position.

(12) Axial motion of the plunger **128** draws fluid from the environment, through the sample strainer section **108** and into the sample volume section **106**. The axial motion continues until the sample volume cap **126** reaches the bottom of the sample volume section **106**. The sample volume cap **126** seals the bottom of the sample volume section **106** and the plunger **128** seals the top of the sample volume section **106**.

(13) In the second axial position, the spring **118** maintains a closing force on the plunger shaft **112**. The desired fluid sample is trapped between the plunger **128** and the sample volume cap **126**. Circumferential seals **136** are provided around the sample volume cap **126** and the plunger **128** to form a watertight seal between the plunger and an interior wall of the sample volume section **106** and between the sample volume cap and the sample volume section in order to prevent leakage from the sample volume. The fluid sample remains inside the sample volume section **106** until extracted. The sample volume cap **126** may be manufactured from elastomer to allow penetration by a syringe to extract fluid from the sample volume section **106**.

(14) FIG. **3** depicts a fluid sampling system **150**. The fluid sampling system includes a housing **152** holding a plurality of sampler modules **100**. The sample strainer section **108** of each of the sampler modules **100** extends into a central chamber **154** of the housing **152**. Using this arrangement, fluid flow through the central channel **154** can be sampled at intervals as triggered by the fusible links **134** of each individual sampler module **100** to collect a multitude of samples, each in a sample storage chamber, at a sample schedule, as needed.

(15) Referring again to FIG. **1**, the housing **152** can include threads **156** so that the sampler modules **100** can threadedly engage with the housing **152**.

(16) The fluid sampling system **150** is pressure balanced such that the system can operate at any ambient pressure without compromise in performance or risk of failure. The fluid sampling system **150** and each sampler module **100** can be scaled to the size of the application; thereby, allowing for very large or small samples to be collected.

(17) The foregoing description of the preferred embodiments of the invention has been presented for purposes of illustration and description only. It is not intended to be exhaustive nor to limit the invention to the precise form disclosed; and obviously many modifications and variations are possible in light of the above teaching. Such modifications and variations that may be apparent to a person skilled in the art are intended to be included within the scope of this invention as defined by the accompanying claims.

Claims

1. An apparatus for obtaining fluid samples comprising: a body having a spring retention section, a sample volume section, and a sample strainer section, wherein the bottom of said spring retention section is attached to the top of said sample volume section and the bottom of said sample volume section is connected to the top of said sample strainer section; a plunger shaft having a first end and a second end installed in said body and extending axially from the top of said spring retention section, through said sample volume section, to the bottom of said sample strainer section; a spring disposed around the first end of said plunger shaft; a sample volume cap attached to the second end of said plunger shaft; and a plunger attached to said plunger shaft with said plunger located between said sample volume section and said sample strainer section.
2. The apparatus of claim 1, wherein the top of said sample volume section is closed and includes an aperture for said plunger shaft to pass therethrough.
3. The apparatus of claim 2, further comprising a disk attached to said plunger shaft wherein said spring rests on the top of said sample volume section and is held in place by said disk.
4. The apparatus of claim 3, further comprising: a J-slot located at the top of said spring retention section; pins protruding from the periphery of said disk into said J-slot; and a fusible link for preventing motion of said pins inside said J-slot.
5. The apparatus of claim 4, wherein electrical current passed through said fusible link causes said fusible link to fail and allows said pins to move inside said J-slot; wherein force from said spring against said disk moves said plunger shaft drawing fluid into said sample volume section, and wherein motion of said plunger shaft continues until said sample volume cap reaches the bottom of said sample volume section.
6. The apparatus of claim 1, wherein said sample volume cap is elastomer.
7. The apparatus of claim 1, further comprising circumferential seals around said sample volume cap and said plunger.
8. The apparatus of claim 7, wherein said circumferential seals are capable of forming a watertight seal between said plunger and an interior wall of the sample volume section and between said sample volume cap and said sample volume section.
9. A fluid sampler module comprising: a tubular body having a top and a bottom; a spring operated plunger shaft extending axially from a top to a bottom of said tubular body; a sample volume cap attached to an end of said spring operated plunger shaft; and a plunger attached to said spring operated plunger shaft; wherein axial motion of said spring operated plunger shaft draws fluid into said tubular body trapping the fluid between said plunger and said sample volume cap inside said tubular body.
10. The fluid sampler module of claim 9, wherein said sample volume cap is manufactured from elastomer.
11. The fluid sampler module of claim 9, wherein said tubular body comprises a spring retention section connected to the top of a sample volume section and a sample strainer section connected to the bottom of the sample volume section.
12. The fluid sampler module of claim 11, wherein the top of the sample volume section is closed and includes an aperture for said spring operated plunger shaft to pass therethrough; and wherein a spring rests on the top of the sample volume section and is held in place by a disk attached to said spring operated plunger shaft.
13. The fluid sampler module of claim 12, further comprising: a J-slot located at the top of the spring retention section; pins protruding from the periphery of the disk into said J-slot; and a fusible link for preventing motion of said pins inside said J-slot.
14. The fluid sampler module of claim 13, wherein electrical current passed through said fusible link causes said fusible link to fail and allows said pins to move inside said J-slot; wherein force

from said spring against said disk moves said spring operated plunger shaft drawing fluid into the sample volume section; and wherein motion of said spring operated plunger shaft continues until said sample volume cap reaches the bottom of the sample volume section.

15. The fluid sampler module of claim 10, further comprising circumferential seals around said sample volume cap and said plunger; and wherein said circumferential seals form a watertight seal between said plunger and an interior wall of the sample volume section and between said sample volume cap and the sample volume section.

16. A fluid sampling system comprising: a housing having a central chamber capable of receiving fluid flow; and a plurality of fluid samplers attached to said housing, each fluid sampler arranged to obtain a fluid sample from said central chamber; wherein each fluid sampler of said plurality of fluid samplers comprises: a body having a spring retention section connected to the top of a sample volume section and a sample strainer section connected to the bottom of the sample volume section, the sample strainer section being disposed in the central chamber; a plunger shaft extending axially from the top of the spring retention section to the bottom of the sample strainer section; a spring disposed around the plunger shaft in the spring retainer section; a sample volume cap attached to an end of the plunger shaft in the sample strainer section; and a plunger attached to the plunger shaft between the sample volume section and the sample strainer section.

17. The fluid sampling system of claim 16, wherein the top of said sample volume section is closed and includes an aperture for the plunger shaft to pass therethrough.

18. The fluid sampling system of claim 17, further comprising a disk attached to said plunger shaft wherein said spring rests on the top of the sample volume section and is held in place by said disk.

19. The fluid sampling system of claim 18, further comprising: a J-slot located at the top of the spring retention section; pins protruding from the periphery of the disk into said J-slot; and a fusible link preventing motion of said pins inside said J-slot.

20. The fluid sampling system of claim 19, wherein electrical current passed through said fusible link is capable of failing to break said fusible link and allowing said pins to move inside said J-slot; wherein force from said spring against said disk moves said plunger shaft drawing fluid into the sample volume section; and wherein motion of the plunger shaft continues until said sample volume cap reaches the bottom of said sample volume section.
